

Cybersecurity & Linux Study Notes

OS basics · Linux commands · Terminal shortcuts · IAM · CIA Triad · EC2 CIA Lab · Tools & Use Cases

Each topic includes a worked example.

1. Operating System (OS) Overview

An **Operating System** is the user interface between the user and the hardware. Any application (software) you use sits on top of the OS. Linux is a popular, open-source, Unix-like OS widely used on servers, cloud (AWS EC2) and in cybersecurity.

Example: When you open a browser, the browser (software) asks the OS to use the network card and memory. The OS handles it for the app.

Linux terminal basics from notes

Command	Meaning	Example
pwd	Print Working Directory: shows where you are	\$ pwd → /home/ubuntu
last	Shows login history / last logins	\$ last → ubuntu pts/0 Mon Sep 29 09:15
w	Shows who is logged in and what they are doing	\$ w
w --help	Shows the help/options for a command	\$ w --help
q	Quit/exit from viewers like less, man	Press q inside man ls
less	View a file page by page	\$ less /var/log/syslog
sudo apt update -y	Refresh package list; -y auto-answers yes ("why" noted: -y = yes)	\$ sudo apt update -y

Example session:

```
$ pwd
/home/ubuntu
$ last | head -3
ubuntu pts/0 203.0.113.5 Mon Sep 29 09:15 still logged in
$ sudo apt update -y
$ w --help
```

2. Linux Navigation & Cursor Movement Shortcuts

Used to move around the command line quickly: move by character, by word, or to line start/end.

Shortcut	Function	Example
Ctrl + A	Beginning of line	Typed <code>sudo apt updat</code> → Ctrl+A jumps to start to fix
Ctrl + E	End of line	Jump back to end after editing
← / →	Move left / right 1 character	Fix a single typo
Ctrl + B / Ctrl + F	Back / forward 1 character	Same as arrow keys
Alt + B / Alt + F	Back / forward 1 word	Skip over words quickly
Ctrl + ← / Ctrl + →	Back / forward one word	Works in many terminals
Home / End	Beginning / end of line	Same as Ctrl+A / Ctrl+E

Example: You typed `sudo apt instal nmap` (typo).

1) Press **Alt+B** twice to go back to “instal”. 2) Press → and type **l**. 3) Press **Ctrl+E** and Enter.

More shortcuts

Shortcut	Function
Ctrl + C	Stop current command
Ctrl + Z	Suspend current process
Ctrl + D	Exit shell
Ctrl + L	Clear screen
Ctrl + U	Delete from cursor to beginning
Ctrl + K	Delete from cursor to end
Ctrl + W	Delete previous word
Alt + D	Delete next word
Ctrl + Y	Paste deleted text
Ctrl + T	Swap two characters
Ctrl + R	Search command history
Ctrl + G	Exit history search
Tab	Auto-complete
↑ / ↓	Previous / next command
!!	Repeat previous command
!\$	Use last argument of previous command
Ctrl+Shift+T / W	New / close terminal tab

Most important: Ctrl+C, Ctrl+L, Ctrl+R, Tab, ↑ and ↓.

Example: Forgot to use sudo? Type `sudo !!` to rerun the last command as admin.

3. Basic Linux Commands (with examples)

Files & directories

Command	Use	Example
pwd	Show current directory	pwd
ls / ls -l / ls -a	List files / detailed / hidden	ls -la
cd	Change directory	cd /var/log cd .. cd ~
mkdir	Create directory	mkdir lab
rmdir	Remove empty directory	rmdir lab
touch	Create empty file	touch notes.txt
cat	Display file	cat notes.txt
less / head / tail	View page-wise / first / last lines	tail -n 20 /var/log/auth.log
cp	Copy	cp notes.txt backup.txt
mv	Move/rename	mv notes.txt old.txt
rm / rm -r	Delete file / directory recursively	rm old.txt
echo	Display text / write to file	echo hello > hi.txt
nano	Text editor	nano hi.txt
grep	Search text	grep 'Failed' /var/log/auth.log
find	Find files	find / -name '*.conf'
clear, date, cal	Clear screen, show date, calendar	date
whoami, hostname, uname	User, computer name, system info	uname -a
man, history	Manual pages, command history	man ls

Permissions & disk

Command	Use	Example
chmod	Change permissions	chmod 600 key.pem
chown / chgrp	Change owner / group	sudo chown ubuntu file
umask	Default permissions	umask
stat, file	File info / type	stat key.pem
du, df	Disk usage / free space	df -h
sudo	Run as administrator	sudo apt update -y

Processes

Command	Use	Example
ps, top, htop	Show / monitor processes	ps aux
kill	Terminate a process	kill 1234
jobs, bg, fg	Manage background jobs	sleep 100 & ; jobs
nohup	Run after logout	nohup ./run.sh &

Networking

Command	Use	Example
ip addr / ip route	IP addresses / routes	ip addr
ping	Test connectivity	ping -c 4 google.com
ss	Network connections	ss -tuln
curl	HTTP requests	curl -I https://example.com
wget	Download files	wget https://example.com/file.zip
dig, nslookup	DNS lookup	dig example.com
traceroute	Trace network path	traceroute example.com

Useful for cybersecurity

Command	Use	Example
sha256sum / md5sum	File hash (integrity)	sha256sum file.iso
strings	Readable text in binaries	strings a.out head
who, w, last	Logged-in users / login history	last -n 5
journalctl	System logs	journalctl -u ssh
dmesg	Kernel messages	dmesg tail
lsof	Open files/connections	sudo lsof -i :22

Package management

Ubuntu/Debian:	Fedora/RHEL:
sudo apt update	sudo dnf update
sudo apt upgrade	sudo dnf install <package>
sudo apt install <package>	
sudo apt remove <package>	

Example: sudo apt install nmap -y

4. IAM: Identity & Access Management

IAM makes sure the right people get the right access to the right resources at the right time. It is a key control for **Confidentiality**.

The “A”s of IAM (from notes)

Term	Meaning	Example
Identity	Who you are: a unique user/account	User account <i>ravi@company.com</i>
Authentication	Proving who you are	Password + MFA code
Authorization	What you are allowed to do	Ravi can read the HR folder, not edit it
Auditing	Reviewing/recording activity	Checking who opened the HR folder last week
Accounting	Tracking usage (logs)	Log: Ravi logged in 09:15, used 2 GB
Administration	Creating, changing, removing users and roles	Onboarding/offboarding an employee
Availability	Access is available when needed	Login service stays up 24/7

Example (login flow): Ravi enters username (identity) → password + OTP (authentication) → system checks his role (authorization) → every action is logged (accounting/auditing).

Where identities come from

Identities come from a directory or identity provider such as **Active Directory (AD)**, **Microsoft Entra ID**, **Okta**, **AWS IAM**, **Google Cloud identity**. The remaining lifecycle and access governance is controlled by **IGA** tools (Identity Governance & Administration) such as **SailPoint** and **Saviynt**. **Privileged Access Management (PAM)** covers admin/privileged accounts.

Layer	Purpose	Tools
Identity Provider	Store identities, do login/SSO	AD, Entra ID, Okta, AWS IAM, Google Cloud
IGA	Access requests, reviews, joiner/mover/leaver	SailPoint, Saviynt
PAM	Protect and monitor admin accounts	CyberArk, BeyondTrust (examples)

Example: A new employee joins → HR record → IGA tool creates the account in Entra ID → assigns roles → manager approves access → admin access goes through PAM with session recording.

5. CIA Triad in Cybersecurity

Principle	Meaning	Common Tools	Use Cases
Confidentiality	Protect information from unauthorized access	Encryption, VPN, Access Control, Password Managers	Protect personal data, passwords, financial info
Integrity	Data is accurate and not improperly modified	Hashing, Digital Signatures, File Integrity Monitoring	Detect altered files, verify software, protect databases
Availability	Systems and data accessible when needed	Backup tools, RAID, Load Balancers, DDoS Protection	Disaster recovery, preventing downtime

Confidentiality: tools and use cases

Tool	Use Case
Encryption	Protect sensitive data in storage and transit
VPN	Secure network communication
Access Control	Only authorized users access data
Passwords & MFA	Prevent unauthorized account access
BitLocker	Encrypt computer storage
OpenSSL	Secure communication using encryption and certificates
PGP/GPG	Encrypt emails and files
IAM	Manage user permissions and access
Firewalls	Block unauthorized network access
Password Managers	Store and manage passwords securely

Examples: protecting student/employee records, securing banking information, encrypting confidential files, securing email, preventing unauthorized database access.

Hands-on examples:

```
# Confidentiality: encrypt a file with GPG
gpg -c secret.txt                # creates secret.txt.gpg

# Integrity: detect modification with a hash
sha256sum report.pdf > report.sha256
sha256sum -c report.sha256       # OK = unchanged, FAILED = modified

# Availability: quick backup + check service is up
tar -czf backup.tar.gz /home/ubuntu/data
systemctl status ssh
```

6. Your EC2 CIA Lab (AWS EC2 – Linux)

Launch one AWS EC2 Linux instance and demonstrate all three CIA pillars on it.

Pillar	Controls	Example on EC2
Confidentiality	IAM / MFA, SSH, permissions, encryption	Log in only with SSH key; <code>chmod 600 key.pem</code> ; enable MFA on IAM user; encrypt EBS volume
Integrity	SHA256, FIM, Git, Audit	<code>sha256sum</code> on files; file integrity monitoring (e.g. AIDE); keep code in Git; <code>auditd</code> / CloudTrail logs
Availability	Backup, Monitoring, Health check, Recovery	EBS snapshots; CloudWatch alarms; load balancer health checks; restore from snapshot / AMI

Sample lab commands

```
# --- Confidentiality ---
chmod 600 mykey.pem
ssh -i mykey.pem ubuntu@<EC2-PUBLIC-IP>
ls -l /home/ubuntu                # check permissions
sudo chmod 640 secret.txt          # owner rw, group r, others none
gpg -c secret.txt                  # encryption

# --- Integrity ---
sha256sum important.txt            # note the hash
echo "tamper" >> important.txt
sha256sum important.txt            # hash now different = modified
git init && git add . && git commit -m "baseline"
sudo apt install auditd -y         # audit logs
sudo ausearch -m USER_LOGIN | tail

# --- Availability ---
sudo systemctl status ssh          # service health
df -h ; top                        # disk and CPU monitoring
tar -czf ~/backup.tar.gz ~/data    # backup
# Recovery: restore with tar -xzf ~/backup.tar.gz
```

7. Cybersecurity Tools & Use Cases

Tool	Purpose	Example
Wireshark	Network traffic analysis	Capture packets to see who talks to whom
Nmap	Network scanning and discovery	<code>nmap -sV 192.168.1.10</code> (on systems you own/are authorized to test)
Burp Suite	Web application security testing	Inspect and test a web login form
Metasploit	Security testing / vulnerability validation	Validate a patched vulnerability in a lab
Kali Linux	Cybersecurity testing platform	Lab OS with security tools pre-installed
Nessus	Vulnerability scanning	Scan servers for missing patches
Snort	Intrusion detection	Alert on suspicious network patterns
SIEM tools	Log monitoring, security-event analysis	Alert on 20 failed logins in 1 minute
BitLocker	Disk encryption	Protect a lost laptop's data
OpenSSL	Encryption and certificates	<code>openssl rand -hex 16</code>

Major use cases

Network security · Vulnerability assessment · Penetration testing · Malware analysis · Web application security · Identity and access management · Data protection · Incident detection and response · Digital forensics · Security monitoring · Phishing detection · Backup and disaster recovery

Note: use scanning/testing tools only on systems you own or have written permission to test.